Md Ashik Khan

aasshhik98@gmail.com • +880 1796 103496 • ashik-khan-cs.github.io
Google Scholar • GitHub • LinkedIn

Research Interests

Computer Vision, Medical Image Analysis, Multimodal Deep Learning, Video Action Recognition, Trustworthy AI

Education

Indian Institute of Technology Kharagpur West Bengal, India
Master of Technology in Computer Science and Engineering *2021 – 2023*

- CGPA: 8.76/10.00 • Advisor: Dr. Abir Das • Scholarship: ICCR (fully funded)
- Thesis: *A Large-scale Study of Representation Learning and the Benchmarking of Video Action Recognition*

Bangladesh Army University of Science and Technology Saidpur, Bangladesh
Bachelor of Science in Computer Science and Engineering *2015 – 2019*

- CGPA: 3.35/4.00 (Final-year GPA: 3.61/4.00)

Publications

Refereed Publications

1. **Md Ashik Khan**, Abu Saleh Musa Miah, Fahmid Al Farid, Md Abdur Rahim, and Hezerul Abdul Karim. "Human Activity Recognition Under Low-Light Conditions Using Frozen CLIP with Lightweight Adaptation." *Discover AI* (Springer Nature), 2026 (Accepted).
2. **Md Ashik Khan** and Md Nahid Siddique. "Fixed-Budget Parameter-Efficient Training with Frozen Encoders Improves Multimodal Chest X-Ray Classification." *Proceedings of the 28th International Conference on Computer and Information Technology (ICCIT 2025)*, Cox's Bazar, Bangladesh, December 2025, pp. 2748-2753. [DOI].
3. **Md Ashik Khan** and Arafat Alam Jion. "Fixed-Threshold Evaluation of a Hybrid CNN-ViT for AI-Generated Image Detection Across Photos and Art." *Proceedings of the 28th International Conference on Computer and Information Technology (ICCIT 2025)*, Cox's Bazar, Bangladesh, December 2025, pp. 3853-3858. [DOI].
4. **Md Ashik Khan** and Rafath Bin Zafar Auvee. "Comparative Analysis of Resource-Efficient CNN Architectures for Brain Tumor Classification." *Proceedings of the 27th International Conference on Computer and Information Technology (ICCIT 2024)*, Cox's Bazar, Bangladesh, December 2024, pp. 639-644. [DOI].

Under Review

- Md Nahid Siddique*, **Md Ashik Khan***, Md Rezaul Karim Khan, Naphtali David Rishe, and Dongsheng Luo. "High Accuracy, Weak Transfer: A Cross-Domain Audit of Fake News Detection." Submitted to *WI-IAT 2026*. (*Equal contribution)
- **Md Ashik Khan** and Md Nahid Siddique. "SHIFT-M3: Pre-fusion Alignment-based Consistency Screening for Multimodal ECG Record Integrity." Submitted to *Machine Learning for Healthcare (MLHC 2026)*.
- **Md Ashik Khan**, Abu Saleh Musa Miah, Md Nahid Siddique, Md Abdur Rahim, and Fahmid Al Farid. "Prompt Sensitivity and Answer-Position Bias in VideoLLM Action Recognition." Submitted to *WACV 2027*.

Ongoing Research

- **Md Ashik Khan**, Abu Saleh Musa Miah, Md Abdur Rahim, and Fahmid Al Farid. "Reliability-Aware Cross-Modal Consistency for Label-Efficient Human Activity Recognition."
- **Md Ashik Khan**, Abu Saleh Musa Miah, Fahmid Al Farid, and Md Abdur Rahim. "WICA-Net: A Lightweight Wavelet-Integrated Coordinate-Attention Network for Generalizable Brain Tumor MRI Analysis."

Research Experience

CVIR Lab, IIT Kharagpur West Bengal, India
Graduate Research Assistant (M.Tech Thesis Research) | Advisor: Dr. Abir Das *Jun 2022 – Apr 2023*

- **Thesis:** *A Large-scale Study of Representation Learning and the Benchmarking of Video Action Recognition*
- Systematically evaluated transferability across 6 SOTA video architectures (SlowOnly, TimeSformer, SIFAR) on 14 downstream benchmarks for self-supervised and supervised representations across 2D/3D CNNs and Video Transformers.

- Created and validated the Construction Action Dataset (1,844 annotated videos across 13 industrial activity categories) to address domain-specific data scarcity in action recognition.
- Mapped optimal transfer pathways under domain-shift and data-scarcity constraints through benchmarking out-of-domain generalization and representation alignment.

Independent Research Collaborations *May 2023 – Present*

Collaborators: Dr. Abu Saleh Musa Miah (Dept. of CSE, University of Rajshahi, Bangladesh) · Dr. Fahmid Al Farid (Faculty of CS & Informatics, BSBI, Germany) · Dr. Md Abdur Rahim (Dept. of CSE, PUST, Bangladesh) · Dr. Hezerul Abdul Karim (FAIE, Multimedia University, Malaysia)

- Designed a parameter-efficient temporal adaptation framework for frozen CLIP, achieving a +6.01% accuracy gain under low-light domain shifts with under 0.2% additional computational overhead. *(Discover AI, Springer Nature, 2026)*
- Evaluated PEFT adapter variants under a fixed parameter budget on frozen multimodal encoders, outperforming full fine-tuning by 13.8 pp AUROC (0.908 vs. 0.770) with 40× fewer parameters. *(ICCIT 2025)*
- Designed WICA-Net by integrating wavelet decomposition with coordinate attention to capture multi-scale frequency features for generalizable brain tumor MRI analysis. *(in preparation)*
- Designed SHIFT-M3, a pre-fusion alignment-based screening method to ensure multimodal ECG record integrity and flag data misalignment under noisy clinical collection conditions. *(submitted, MLHC 2026)*
- Developed RACM, a semi-supervised skeleton-IMU HAR framework using inter-branch Jensen-Shannon agreement as a training signal, raising missing-skeleton accuracy from 23.6% to 45.2% (+21.6 pp) and exceeding all baselines by >10 pp on both missing-modality metrics under 25% labeled data. *(in preparation)*
- Conducted a controlled response-regime audit of 5 VideoLLMs across 8 action-recognition datasets under 3 interfaces, uncovering 23–96 pp accuracy swings on identical clips from prompt changes alone. *(submitted, WACV 2027)*

Selected Projects

Occlusion-Robust Pedestrian Detection with YOLOv12-S and Transformer Backbones *Autumn 2025*

- Benchmarked YOLOv12-S (9.25M parameters, 21.5 GFLOPs) with Swin Transformer and ConvNeXt backbones on EuroCity, comparing global context modeling against local feature extraction for pedestrian detection.
- Designed a stratified occlusion robustness evaluation across four levels (none, light, moderate, heavy) using occlusion-aware dataset variants with per-image metadata.
- Achieved mAP50 ≈ 75% and precision ≈ 92% overall; heavy-occlusion mAP50 of 75.3% demonstrated Swin Transformer backbone resilience under severe partial visibility.

Deep Learning-based Hidden Camera Generalization Study *Spring 2024*

- Synthesized a domain-randomized dataset of surveillance camera placements to address real-world data scarcity in covert camera detection.
- Fine-tuned YOLOv8 and ResNet50 with custom augmentation pipelines to analyze generalization under heavy occlusion and low-resolution conditions, achieving 92% mAP.

Context-Specific Quote Recommendation from Historical Text *Autumn 2022*
IIT Kharagpur

- Built a context-based quote recommendation system using Hugging Face DistilBERT, applying full fine-tuning on the Quotation POTUS dataset for domain-specific semantic text retrieval.
- Achieved 82.25% accuracy on quote phrase matching, validating the effectiveness of transformer-based transfer learning on domain-specific historical text.

Professional Experience

Auptimate Dhaka, Bangladesh
Senior Software Engineer *Jan 2026 – Present*
Software Engineer *May 2023 – Dec 2025*

- Architected a scalable e-signature and document automation platform with microservices (FastAPI, React.js, PostgreSQL, AWS), supporting end-to-end digital onboarding for enterprise clients.
- Engineered an NLP-powered AI agent for automated legal document processing and semantic query resolution.

Hovata Technologies Dhaka, Bangladesh
Assistant Programmer *Mar 2020 – Aug 2021*

- Built and maintained PetrolERP for petroleum business management (sales, stock, expenses, vendors), implementing real-time inventory tracking that reduced stock discrepancies by 35% via transaction locks.

- Led development of parking management system optimizing space utilization using React.js, Redux, PHP Lumen, and MySQL.

Teaching Experience

Graduate Teaching Assistant | **Department of CSE, IIT Kharagpur**

- **Software Engineering** *Spring 2022*
- **Algorithms-I** *Autumn 2022*

Responsibilities: Conducted tutorials, evaluated assignments, mentored 20 students during laboratory sessions, and supervised multiple groups during semester projects.

Academic Service & Certifications

- **Journal Reviewer:** *IEEE Access*
- **Fundamentals of Accelerated Data Science** • NVIDIA Deep Learning Institute *2021*

Scholarships & Awards

- **ICCR Scholarship for M.Tech at IIT Kharagpur** • Fully Funded international scholarship *2021 – 2023*
- **Finalist** • BAUST-IEEE Idea Contest • BAUST *2018*
- **Honorable Mention** • Inter-University Programming Contest • IUBAT, Dhaka *2016*
- **Board Talent Rank 35th** • Secondary School Certificate (SSC) Examination, Dhaka Board *2012*

Technical Skills

Languages: Python, C++, SQL
AI & Vision: PyTorch, OpenCV, MMAction2, YOLO
Tools: Git, Linux, LaTeX, AWS